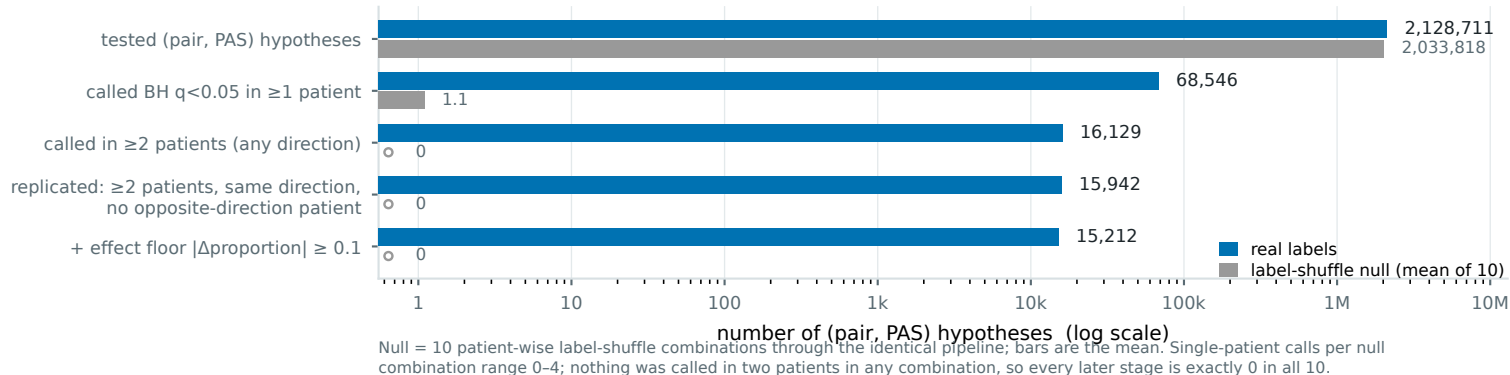


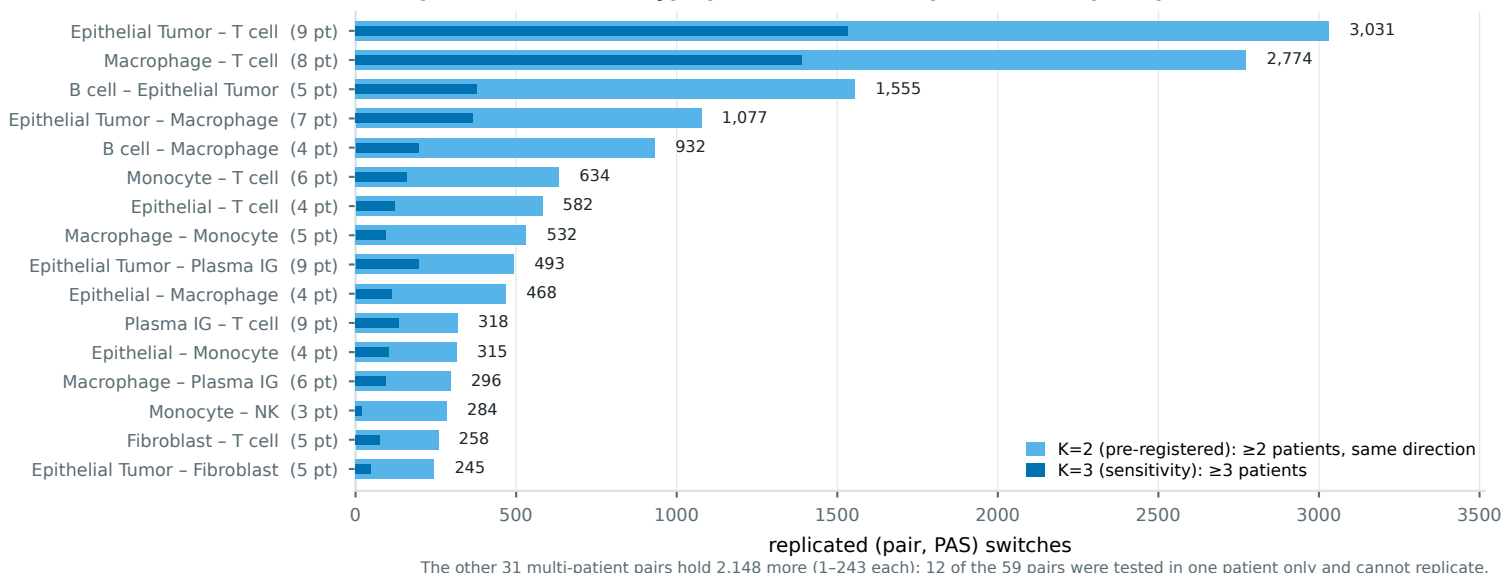
Fig 6 | Reliable cell-type APA switches in a tumour cohort

Laughney lung adenocarcinoma, 12 patients (15 scRNA-seq libraries): 15,942 of 2,128,711 tested (pair, PAS) hypotheses replicate in ≥ 2 patients in the same direction — and none replicate in any of the 10 patient-wise label-shuffle nulls.

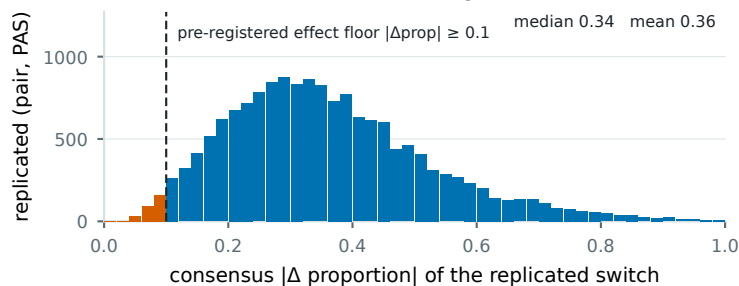
a Replication funnel, real vs patient-wise label-shuffle null (one axis)



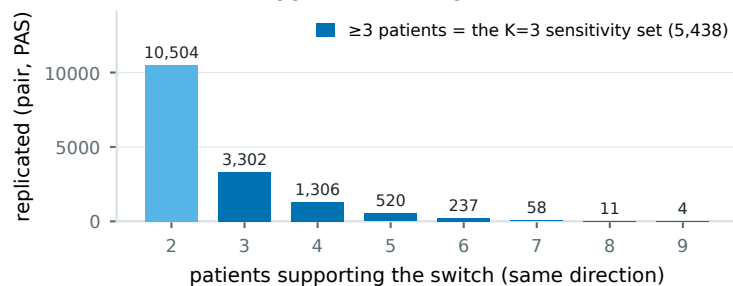
b Top 16 of the 47 cell-type pairs tested in ≥ 2 patients ('n pt' = patients tested)



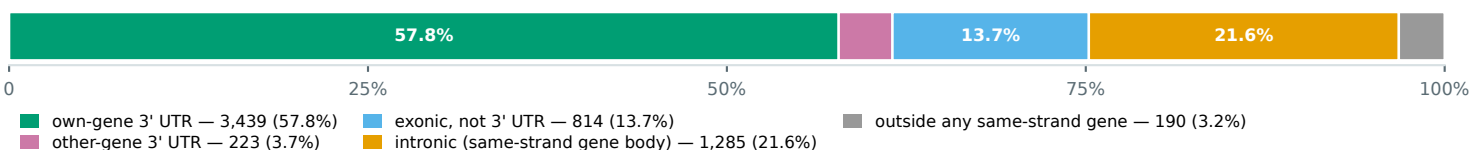
c Effect size of the 15,942 replicated switches



d Patient support of the replicated set



e Genomic context of the 5,951 replicated PAS (recomputed here from the Ensembl GRCh38.99 GTF)



Exclusive, strand-matched partition of the distinct replicated PAS. Roll-ups: 96.8% of the replicated PAS sit inside a same-strand gene body, 75.2% are exonic, 61.5% are in some gene's 3' UTR, 21.6% are intronic and 3.2% fall outside any same-strand gene (20 disclosure 1 quotes 61.5%, 57.8% and 96.8%, all reproduced here; the rest are recomputed for this figure). BUT only 57.8% lie in the 3' UTR of the gene the caller ASSIGNED them to: 223 of the 3,662 3'-UTR PAS (6.1%) lie in a DIFFERENT gene's 3' UTR, and 544 (9.1%) lie inside ≥ 2 overlapping same-strand gene models. NO ranked list of named top-switch genes is shown here: 12 of the top-30 gene-level rows in 20 (40%) name a gene whose 3' UTR does not hold the PAS, 8 of them a different gene's 3' UTR, and seven have zero assigned PAS cohort-wide (tool issue #99). Replication statistics are unaffected — they are computed on PAS identifiers, never on gene names.

Design: one cohort run so all 15 libraries share a single PAS identifier space; pre-registered precision-first universe (IP-filtered $\wedge$ tier-1 $\wedge$ cohort clip-molecule sum ≥ 2) = 80,464 PAS; 29,063 curated cells $\rightarrow$ 18,651 confirmed (64.2%) by the pre-registered label policy (20), of which 17,373 entered the tests in the 12 primary patients; 59 cell-type pairs; test = Fisher on cells with no marker pre-selection (14); unit of replication = patient, so the 3 patients with 2 libraries each count once. RECORD: PeakATail code 9dfdefb (#96 minus-strand IP fix) — v2-code record, verified 2026-08-22; supersedes v1-code 4efeb125, kept in 20 as the labelled comparison row; source of truth manuscript/20_stage3_replication.md (verifier verdict SOUND). NULL: not one feature replicates in ANY of the 10 patient-wise label-shuffle combinations run through the identical pipeline: 0 replicated in every one of the 10, over a comparable 2,033,818 (pair, PAS) hypotheses tested per combination (58.9M null tests produced 11 nominal $q < 0.05$ calls, none co-occurring in two patients). Ten permutations resolve only to an empirical $p \leq 0.091$; report this as 'none in 10 nulls', NEVER as an FDR estimate. EXCLUSION: GSM3516664-MetBone is excluded by pre-registration (13 addendum item 8), but its stated premise was WRONG — the 0.015% clip rate came from the caller sampling only the first 200,000 CB reads of a coordinate-sorted BAM (head of chr1); MetBone in fact has 306k clip molecules (tool issue #99). Never repeat the 'no clip evidence' justification. The 13-patient sensitivity run that includes it gives 16,010 / 15,282, a 0.4% difference confined to one pair. GENE NAMES: no named top-gene list may be published until PAS $\rightarrow$ gene re-assignment (panel e; 20 disclosure 1, tool issue #99) — the replication statistics themselves are computed on PAS ids. CAVEATS: one cohort, one chemistry, one caller — no orthogonal 3'-end assay confirms these sites; pairs tested in more patients replicate more, so panel b tracks cohort composition as much as biology; one library (GSM3516672) trips the per-GSM null rule only through BH discreteness (3 hits in 9.1M null tests) and is kept and disclosed (20). Panel c plots the consensus effect, but the pre-registered floor is applied PER PATIENT and the patients re-counted, so it removes 730 switches (15,942 $\rightarrow$ 15,212) — more than the 291 whose consensus alone falls below 0.1. Every plotted value: results/figures/manuscript/laughney_switches*.tsv.